Speed, distance and time

1 If the average speed of a journey was 63 km/h, determine whether the following statements are true or false:

a The minimum speed of the journey was 63 km/h.

b The maximum speed over the journey was 63 km/h.

c 63 km were travelled in exactly 1 hour.

d The speed could have waivered above and below the speed of 63 km/h.

e The journey took 1 hour.

2 Maria travels by car for 420 km. The trip takes 10 hours. What is the average speed of the trip?

3 Amelia travelled by car, crossing a distance of 340 km in 5 hours. At what average speed was the car travelling?

4 Sally tells Neil that her trip from Uppity to Downtown had an average speed of 90 km/h.

a If the trip took 3 hours, what distance was covered?

b State another example of how long the the trip could have taken and what distance would have been covered at an average speed of 90 km/h.

5 A dog travels 42 km in 1.4 hours. At what speed is it travelling?

6 A man drives a truck 144 km in 2 hours, then stops to refuel and eat lunch at a petrol station for an hour. He then drives the truck for another 5 hours, covering 96 km. What was the truck's average speed throughout the whole journey, in kilometres per hour?

7 At 11:45 am, Christa and Sharon leave in their car headed to Brisbane 96 km away. They arrive at 1:15 pm. What is the average speed of the journey?

8 At 9:50 am, Skye and Isabelle leave in their car headed to Melbourne 399 km away. They arrive at 2:35 pm. What was the average speed of the journey?

9 A bus leaves the depot at 6:00 am and 324.5 km later arrives at 11:30 am. What was the average speed of the journey?

10 Two animals were tracked and their movements were measured over different time periods. The cat travelled 31.2 km in 1.4 hours, while the sugar glider possum travelled 95.8 km in 4.2 hours. Which animal travelled more slowly?

11 If a galapagos turtle travels 0.306 km at a s[illegible] of 0.09 km/h, how long will it take the animal to cross the whole distance?

12 If a sugar glider possum travels 6 km at a speed of 15 km/h, how long will it take the animal to cross the whole distance?

13 Luke runs to school. The school is 7.5 km away and he runs at 10 km/h.

- **a** How long will it take Luke to reach school?
- **b** If school starts at 8:55 am, what is the latest time he should leave home?

14 A woman runs 2.4 km at a speed of 8 m/s, then rests for 15 minutes. She then drives to the gym 15 km away at a speed of 75 km/h.

- **a** Calculate the time she takes to complete her run (excluding rest time).
- **b** Find the time it takes for her to drive to the gym.
- **c** What is the total time of the journey?

15 If a lioness travels for 1.5 hours at speed of 42 km/h, how far will the lioness travel?

16 If a cheetah travels for 0.3 hours at a speed of 37 km/h, how far will the cheetah travel?

17 Two animals were tracked and their movements were measured over different time periods. The cat travelled 10 km in 0.1 hours, while the rabbit travelled 34 km in 0.3 hours. Continuing at the rate measured, how far would the cat travel in 0.3 hours?

18 James and Valentina left the airport at the same time and immediately started travelling in opposite directions on the freeway, with James travelling at 42 km/h and Valentina travelling at 33 km/h. How far apart were they after 2.2 hours?

19 If a bike rode 360 m over 45 seconds, find its speed in metres per second, correct to one decimal place.

20 If a scooter travelled at 29 km/h and has covered 139.2 km, how long has it been travelling? Round your answer to one decimal place.

21 If a car travelled at 30 m/s for 19 minutes, find the distance it has travelled in kilometres, correct to one decimal place.

22 If a moped travelled 150 km over 5 hours, find its speed in kilometres per hour.

minutes?

24 If a train travelled at $85 \mathrm{~km/h}$ for 5.4 hours, find the distance it covered in kilometres.

25 Tobias left the airport heading west at $44 \mathrm{~km/h}$. At the same time, Katrina left the airport heading east. After 2.6 hours, Tobias and Katrina were $202.8 \mathrm{~km}$ apart.

- **a** Determine the distance that Tobias had travelled.
- **b** What was Katrina's average speed?

26 Aoife and Ned both left the office and started travelling directly toward the store, but Ned left 3 hours after Aoife, travelling at $88 \mathrm{~km/h}$. Four hours after Ned had left the office, both of them arrived at the store at the same time.

- **a** How many kilometres did Ned travel to reach the store?
- **b** What was Aoife's average speed throughout her journey? Round your answer to one decimal place.

27 Robert made a trip from his house to the beach and back. The trip there took 3 hours and the trip back took 4 hours. He averaged a speed of $36 \mathrm{~km/h}$ on the return trip.

- **a** How far did Robert travel one way from the beach back to his house?
- **b** Given that he took the same route to and from the beach, what was Robert's average speed on the trip from his house to the beach? Round your answer to one decimal place.

28 Ray left home and drove his scooter toward the beach at an average speed of $27 \mathrm{~km/h}$. Some time later, Amelia left, driving her car from the same origin and in the same direction but at an average speed of $56 \mathrm{~km/h}$. After driving for $\frac{18}{5}$ hours, Amelia caught up with Ray. How far did Ray drive before Amelia caught up?

29 Sandy left home and rode her racing bike toward the mountains at an average speed of $32 \mathrm{~km/h}$. Six hours later, Aaron left, driving a scooter from the same origin and in the same direction but at an average speed of $80 \mathrm{~km/h}$. Find the number of hours it took Aaron to catch up with Sandy.

30 Carl and Aicha left different places at the same time and are travelling in the same direction. Carl travelled at an average speed of $21 \mathrm{~km/h}$. Aicha travelled at an average speed of $64 \mathrm{~km/h}$.

- **a** If Carl travelled $231 \mathrm{~km}$ before Aicha caught up with him, find t, the time in hours that it took Aicha to overtake Carl.
- **b** Find d, the additional distance in kilometres, that Aicha had travelled compared to Carl.

31 A cargo plane left the airport and flew towards Newcastle at an average speed of 306 km/h. The plane then flew again towards Brisbane at a speed of 298 km/h. The plane travelled 804.6 km on the second trip.

- **a** How long did it take the cargo plane to travel the second trip from Newcastle to Brisbane?
- **b** Assuming that it took the cargo plane the same time to travel on both trips, what distance did the plane travel from its initial position to Newcastle?

32 Christa made a trip to her friend's house and back. The trip there took 2.7 hours and the trip back took 2.9 hours. She averaged a speed of 80 km/h over the entire return trip there and back.

- **a** How many kilometres in total did Christa travel in the going and return trip?
- **b** If the trip from her friend's house back to home was 215.04 km long, how far was the trip going from home to her friend's house?

33 The distance from Melbourne to Sydney by road is approximately 880 km. How much longer will it take a bus travelling at an average speed of 80 km/h, compared to a car travelling at an average of 95 km/h? Give your answer in hours, correct to two decimal places.

34 The distance from Quiana's house to her work in the city is 40 km. In the morning traffic, she can travel by car at an average speed of 48 km/h or by bicycle at an average of 25 km/h. How much time will she save on the morning commute to work if she uses her car? Give your answer in hours, correct to two decimal places.

35 The distance from Oliver's house to university is 8 km. He can travel by bicycle at an average of 20 km/h or tram at 16 km/h. How much longer will the journey take if he decides to take the tram?

Distance tables

36 The table below shows the distances, in kilometres, between various locations in Australia:

	Brisbane	Sydney	Canberra	Melbourne	Adelaide	Perth
Brisbane	0	1010	1268	1669	3100	4384
Sydney	1010	0	288	963	1427	4110
Canberra	1268	288	0	647	1204	3911
Melbourne	1669	963	647	0	728	3430
Adelaide	3100	1427	1204	728	0	2725
Perth	4384	4110	3911	3430	2725	0

a Georgia, who lives in Brisbane, wants to travel around Australia by bike, which has an average speed of 20 km/h. Which city would be quickest to travel to first, and how how many hours would it take for Georgia to get there?

b If a car can travel at an average speed of 98 km/h, how many hours will it take to travel from Brisbane to Perth? Round your answer to two decimal places.

c Derek wants to travel from Adelaide to Perth.

i How long would it take him to make the trip by bus, if it travels at an average speed of 78 km/h? Round your answer to two decimal places.

ii How long would it take him to make the trip by train, if it travels at an average speed of 237 km/h? Round your answer to two decimal places.

iii How long would it take him to make the trip by car, if it travels at an average speed of 96 km/h? Round your answer to two decimal places.

d Which trip is quicker, Canberra to Brisbane via train at an average speed of 209 km/h, or Canberra to Perth via bus at an average speed of 80 km/h?

37 The table below shows the distances between various locations in Europe:

	London	Paris	Rome	Madrid	Berlin	Athens
London	0	456	841	1715	1099	3151
Paris	456	0	1431	1271	1055	2915
Rome	841	1431	0	1963	1519	1349
Madrid	1715	1271	1963	0	2322	3290
Berlin	1099	1055	1519	2322	0	2346
Athens	3151	2915	1349	3290	2346	0

a If a plane can travel at an average speed of 999 km/h, how long will it take to travel between Berlin and Rome in hours and whole minutes?

b How long does it take to travel from Paris to Madrid via bus then Madrid to Athens via bike, if a bus has an average speed of 80 km/h and a bike has an average speed of 31 km/h? Give your answer in hours and whole minutes.

c How many hours, to two decimal places, would it take to travel from Athens to Madrid via bus at an average speed of 77 km/h?

d How many hours, to two decimal places, would it take to travel from Athens to Paris via bike at an average speed of 22 km/h?

e Which trip is quicker, Athens to Paris via bike or Athens to Madrid via bus?

f How many whole minutes would it take to travel from Rome to Madrid via plane travelling at an average speed of 997 km/h?

g How many whole minutes would it take to travel from Rome to Berlin via bus travelling at an average speed of 70 km/h?

h Which trip is quicker, Rome to Berlin via bus or Rome to Madrid via plane?

38 The table below shows the distances between cities in kilometres:

	Berlin	London	Copenhagen	Amsterdam	Moscow	Rome
Berlin	0	587	613	354	2133	1301
London	587	0	362	925	1601	1175
Copenhagen	613	362	0	950	1562	1526
Amsterdam	354	925	950	0	2479	1433
Moscow	2133	1601	1562	2479	0	2354
Rome	1301	1175	1526	1433	2354	0

Dave states that his trip between Amsterdam and Moscow had an average speed of 80 km/h. Find how long the trip from Amsterdam to Moscow took Dave. Round your answer correct to two decimal places.

39 The table displays the distances between two [redacted]s in kilometres. For example, the distance between Alice Springs and Cairns is 2727 km.

Alice Springs							
3012	Brisbane						
2324	1717	Cairns					
1511	3415	2727	Darwin				
2270	1674	3054	3781	Melbourne			
3630	4384	5954	4045	3452	Perth		
2644	996	2546	4000	868	4144	Sydney	
2096	1467	374	2556	2857	5728	2494	Townsville

Sally states that her trip between Brisbane and Townsville had an average speed of 91 km/h. How many hours did it take Sally to travel from Brisbane to Townsville? Round your answer correct to two decimal places.

40 Emma is taking part in a youth group tour and travels from Amsterdam, departing at 10:45 am, to Prague via Berlin. The first part of the trip took 8.41 hours, and she travelled at the same average speed for the second part of the trip. The distances between towns are shown in the table:

Town A	Town B	Distance from Town A to Town B	Town C	Distance from Town B to Town C
Amsterdam	Berlin	690 km	Prague	380 km
Brussels	Frankfurt	405 km	Geneva	585 km
Grenoble	Milan	390 km	Munich	550 km
Florence	Munich	650 km	Prague	360 km
Paris	Geneva	550 km	Frankfurt	585 km

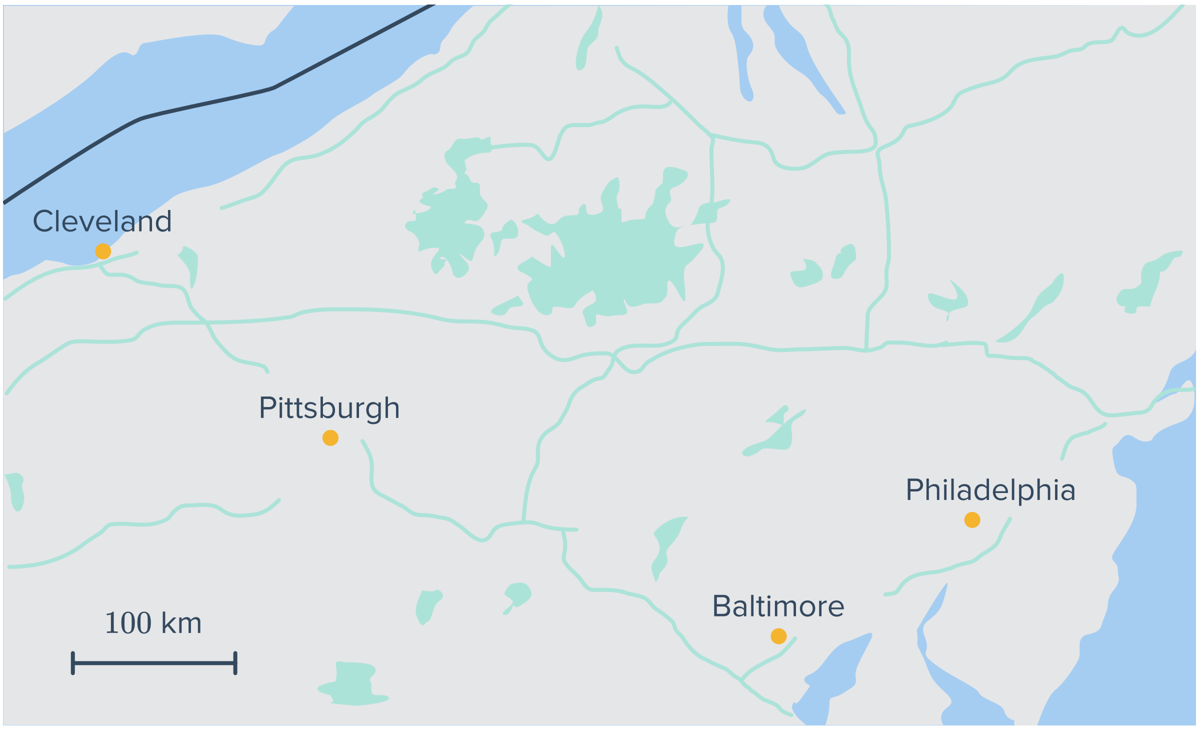
a What was her average speed on the first part of her trip from Amsterdam to Berlin? Round your answer to two decimal places.

b Hence, find the number of hours it took to cover the second part of the trip between Berlin and Prague. Round your answer to two decimal places.

c Determine the time at which Emma arrives at Prague.

Map scales

41 Vincent took a car trip from Pittsburgh to Baltimore. The driving time was 2.9 hours. If the distance between Pittsburgh and Baltimore is 2.8 units as it appears on the map, what is the average speed of the car? Round your answer to two decimal places.



42 Ned took a plane from Brisbane to Townsville. The flight time was 2.4 hours. If the distance between Brisbane and Townsville is 4 units as it appears on the map, what is the average speed of the airplane? Round your answer to two decimal places.

43 Harry took a car trip from Tampere to Kotka. The driving time was 2.5 hours. If the distance between Tampere and Kotka is 1.8 units as it appears on the map, what is the average speed of the car?

44 Justin took an airplane from Lusaka to Durban. The flight time was 3.3 hours. If the distance between Lusaka and Durban is 2.7 units as it appears on the map, what is the average speed of the airplane? Round the answer to the nearest whole number.

45 Abdul is driving from Perth to Coral Bay:

a Given the scale on the map, estimate the distance from Perth to Coral Bay correct to the nearest 100 km.

b If fuel costs $\$1.35$ per litre and the car consumes 13 litres per 100 km, estimate the cost of fuel for the drive.

c If the car travels at an average of 100 km/h, how long will the trip take in hours?

46 Consider the following map:

a Given the scale on the map, estimate the distance from Adelaide to Uluru correct to the nearest 100 km.

b If fuel costs $\$1.40$ per litre and the car consumes 12 litres per 100 km, estimate the cost of fuel for the trip in dollars.

c If the car travels at an average of 95 km/h, how long will the trip take in hours? Round your answer to two decimal places.

47 Consider the following map:

a Given the scale on the map, estimate the distance from Melbourne Airport to the Melbourne CBD correct to the nearest 10 km.

b If a taxi charges $\$1.80$ per kilometre in addition to a $\$3$ booking fee and $\$5$ for road tolls, estimate a fare into town in dollars.

c If the car travels at an average of 60 km/h, how long will the trip take in minutes?

48 Consider the following map:

a Given the scale on the map, estimate the distance from Southbank helipad to Geelong correct to the nearest 10 km.

b A chartered helicopter travels at an average speed of 210 km/h. How many minutes should the flight last?

c The helicopter cost $\$10$ per minute plus a booking fee of $\$60$. Estimate the cost of the chartered flight

49 Aashish wanted to travel from Auckland t[redacted]ington. However, he could not drive the entire distance in one go. He decided to make one stop on the way at a hotel in Taupo.

a It took Aashish 5.4 hours to reach the hotel. If the distance between Auckland and Taupo is $3\ \mathrm{cm}$ on the map, what was his average speed for the first part of the trip? Round your answer to one decimal place.

b Aashish's average speed on the second trip was $53.8\ \mathrm{km/h}$. If the trip took him 8 hours, what is the distance between Taupo and Wellington?

c How far should the distance between Taupo and Wellington appear on the map? Round your answer to one decimal place.

d The room at the hotel cost $\$80$. If the fuel cost is 158 cents/litre, and Aashish's car fuel usage is 7 kilometre/litre, what was the cost of the entire trip from Auckland to Wellington?

50 A taxi company wants to find the cost of a[redacted]t trip from Birkdale to Albany Creek to publish the fee for customers.

a If the route distance between Birkdale and Albany Creek appears to be $8\ \text{cm}$ on the map, what is the actual route distance?

b If the taxi's average speed for the trip is $22\ \text{km/h}$, how many hours does the trip take? Round your answer to one decimal place.

c The taxi driver is paid $\$22.82$ per hour. The car's fuel usage is $8\ \text{km/L}$ and the fuel cost is 147c/L. What is the total cost of the trip?

d If the taxi company wants to publish the fee of the trip for 230% of its total cost, How much would the customer have to pay?

51 A bus company is working out the cost of tickets based on $\$0.31$ per kilometre travelled.

a The following table displays the distances between Huntleys Point and the rest of the areas as they appear on the map. Complete the table below:

Area	Distance on map (units)	Actual distance (km)
Wolli Creek	8	
Kogarah	9.5	
Randwick	6.5	
Maroubra	11.5	

b If a customer is at a station in Huntleys Point and has only $\$5.45$ in his pockets, which areas can he reach?

52 Carl wanted to fly from Perth to Adelaide w[illegible]n overnight stop in Sydney.

a Carl's first plane took 4 hours of flight time. If the airplane was travelling at an average speed of 825 km/h, what should the distance between Perth and Sydney appear as on the map, measured in scaled units?

b Carl's second trip took 2.2 hours. If the distance between Sydney and Adelaide appears to be 2.2 units on the map, what was the average speed of the airplane?

c Carl's overnight stay at a hotel in Sydney costs $\$120$. If the flight company charges 11 cents/km for the flight, what was the total cost of the trip?